Ultra High Risk Pregnancy (UHRP) Care Dashboard

Standard Operating Procedure (SOP), Clinical Algorithms & Technical Architecture

DISTRICT: West Nimar (Khargone), MP	**DEVELOPER:** Dr. Vikar Saiyad (Sankalp)	**DATE:** 26-10-2026
TOTAL ANC: Active District Registry	**INTERVENTION:** Ultra High Risk Pregnancy (UHRP)	**EDITION:** Version 4.0 (Enhanced SOP)

1. Executive Overview & Maternal Health Mission

The Ultra High Risk Pregnancy (UHRP) Care Dashboard is an operational decision-support command center deployed across the 9 administrative blocks of West Nimar (Khargone) district, Madhya Pradesh (Barwaha, Bhagwanpura, Bhikangaon, Gogaon, Jhiranya, Kasrawad, Khargone, Maheshwar, and Segaon). Its objective is the systematic reduction of maternal mortality and neonatal deaths by identifying pregnant women with clinical red flags, tracking longitudinal hemoglobin and blood pressure trajectories across 12 Antenatal Care (ANC) visits, ensuring timely parenteral iron / antihypertensive therapies, and orchestrating planned institutional deliveries.

Funnel Tier	Target Cohort	Clinical Criteria	Operational Action / Treatment Focus
Total ANC Registered	District Registry	Universal 1st Trimester Booking	Universal screening, IFA supplementation, baseline Hb & BP recording
High Risk Flagged (HRP)	Identified at Subcenter	Screened by ANM / CHO	Routine ANC + PMSMA referral on 9th of each month
Ultra HRP Master Cohort	**Active UHRP Line List**	**Critical Red Flag Conditions**	Intensive 12-visit tracking, specialist MO exam, and planned delivery
Severe Anemia (< 7.0 g/dL)	Critical Anemia	Hb < 7.0 g/dL at any visit	Immediate FRU referral: Inj. FCM 1000mg, IV Iron Sucrose, or PRBC
Moderate Anemia (7.0-9.9)	Parenteral Iron Cohort	Hb 7.0 – 9.9 g/dL	Prompt IV Iron Sucrose / FCM initiation before 3rd trimester labor
PIH (Hypertension)	Hypertensive Disorders	BP ≥ 140/90 mmHg	Tab Labetalol 100mg BD / Methyldopa, urine albumin, pre-eclampsia alert
Delivery Overdue	**Past EDD Cohort**	**EDD < Today & Del Pending**	Urgent facility referral, NST, biophysical profile, induction of labor

2. Clinical Protocols & Triage Classification Algorithms

The system evaluates patient risk through standardized clinical rules aligned with National Health Mission (NHM) and Anemia Mukt Bharat guidelines:

A. Anemia Staging & Parenteral Iron Rules (Severe & Moderate)

- **Resolved Anemia (≥ 11.0 g/dL):** Initial presentation of severe/moderate anemia resolved to normal range on latest visit.
- **Mild Anemia (10.0 – 10.9 g/dL):** Transitional stage; maintain therapeutic oral iron (2 IFA tablets daily).
- **Moderate Anemia (7.0 – 9.9 g/dL):** High clinical risk of entering labor with depleted iron stores. Requires prompt initiation of IV Iron Sucrose (100mg elemental iron per dose, alternate days) or Inj. Ferric Carboxymaltose (FCM 1000mg single infusion).
- **Severe Anemia (< 7.0 g/dL):** Critical obstetric emergency. Requires immediate referral to First Referral Unit (FRU) or District Hospital Khargone for parenteral iron therapy or packed red blood cell (PRBC) transfusion.

B. Delivery Overdue Triage & Gestational Age Calculation

- **Delivery Overdue (Past EDD):** Identified when p.del === 0 and the calculated Expected Date of Delivery (EDD) is earlier than the dashboard generation date (or today). *Maternal death and abortion cases are strictly excluded from the overdue list.* Displayed with a prominent pulsing rose badge (■ Delivery Overdue) and dedicated KPI filtering card. Overdue mothers require urgent tracking to rule out post-maturity complications and stillbirth.
- **Maternal Death Gestational Age:** For maternal death cases (extracted from p.mat), gestational age is calculated precisely using Last Menstrual Period (LMP) and the documented Date of Death (rather than today's date), preserving accurate epidemiological records.
- **Abortion Cases Gestational Age:** Since exact dates of abortion are not available, gestational age is frozen at the completed weeks of pregnancy corresponding to the mother's last recorded ANC visit (or registration date), preventing false advancement or overdue flags.

C. Attendance & Doctor Exam Alerts

- **Missed 2+ Consecutive ANC Visits:** Triggered when either (a) the elapsed time since the last visit exceeds 70 days, OR (b) two or more consecutive scheduled visits have no attendance date.
- **Not Seen by Medical Officer (MO):** High-risk pregnancy has attended visits exclusively conducted by ANM / CHO without clinical examination by a Medical Officer or Gynecologist.

D. Pregnancy-Induced Hypertension (PIH) & Pre-Eclampsia

- **Controlled BP (< 140/90 mmHg):** Patient on antihypertensive therapy with normalized blood pressure.
- **Uncontrolled BP (≥ 140/90 mmHg):** Blood pressure remains hypertensive; medication dosage adjustment required.
- **Pre-Eclampsia Alert:** Systolic BP ≥ 140 mmHg or Diastolic BP ≥ 90 mmHg concurrent with Urine Albumin ≥ +2. Requires emergency hospitalization and Pritchard regimen (Inj. MgSO4).

E. Protocols & Frontline Worker Counselling (■■■ ■■■■ ■■■■■■■ ■■■■■)

Dedicated protocols and standardized Hindi counselling points are embedded across all 9 risk factor sections in the dashboard for direct field guidance during home visits:

- **Severe & Moderate Anemia:** High iron diet (Moringa/■■■■, spinach, jaggery, roasted chickpeas, sprouted pulses), avoidance of tea/coffee with meals, Vitamin C fruit intake (guava, amla), 108 emergency transport awareness.
- **PIH & Pre-Eclampsia:** Warning signs (severe headache, blurred vision, epigastric pain, sudden facial edema), low-sodium diet, left lateral rest position, mandatory BP check every 7 days.
- **Sickle Cell & GDM:** Regular hydration, avoidance of cold stress, Folic Acid 5mg daily, nutritional management (MNT), and scheduled glucose monitoring.

3. Mobile-First Field Popups & Direct Communication Architecture

For frontline ANMs, CHOs, and Block Medical Officers reviewing cases on mobile smartphones, the dashboard provides streamlined, minimal modal popups with zero information redundancy:

- **12 ANC Visit Record Popup:** Displays the 12 ANC Visit Record Timeline Tracker prominently on top, followed by a unified Subcenter header above ANM & CHO contact chips with direct one-touch calling (tel: protocol), and clear Medical Officer / attendance status alerts.
- **Clinical Management & Vitals Popup:** Details of management done are displayed on top, featuring active treatment badges (Inj. FCM, Iron Sucrose doses, Antihypertensive therapy, or GDM regimen), admission vitals (for delivered cases), identified risk factors, delivery status, and frontline worker phone numbers.
- **Color-Coded Status Cards:** The management button and popup container dynamically adopt green theme (Done / Prescribed) or orange theme (Action Pending), enabling instant triage during field monitoring visits.

4. Data Ingestion Pipeline & Field Extraction Mappings

Data synchronization is executed by update_dashboard.py. The script ingests raw Excel reports exported from the state pregnancy tracking software and builds the global data structure in data.js:

Target Field	Source Folder / Excel Column	Transform / Normalization Logic
id (MP ID)	HRP Line list (Col 0 / 'MP ID')	String cast, whitespace stripped, primary key
hb_v (12 Hb)	HRP Line list (Cols 15, 19, 23...)	Float parse, invalid non-numerics mapped to null
bp_v (12 BP)	HRP Line list (Cols 16, 20, 24...)	Parsed as 'SYS/DIA', whitespace stripped
v_dates (Dates)	HRP Line list (Cols 14, 18, 22...)	Normalized to 'YYYY-MM-DD' format
mo_v (Provider)	HRP Line list (Cols 17, 21, 25...)	Classified as MO/Specialist vs ANM/CHO/FLW
is_doses	HRP Line list (Cols BA to BF)	List of dates where Iron Sucrose injection was given
fcm / bt	HRP Line list (Cols BG, BH, BI)	Extracted as 'Yes'/'No', facility name, and unit counts
mat (Death)	HRP Line list (Maternal Death col)	Date of death extracted from string Yes (DD-MM-YYYY)
del_cat	Delivery linelist / HRP Line list	Reclassified into DH, SDH, CHC, PHC, Private, Home, In Transit

5. UI Architecture & Responsive Navigation System

The interface incorporates Apple iOS Liquid Glass design aesthetics, featuring frosted glass panels (backdrop-filter: blur(20px)), subtle borders, high-contrast typography, and a 4-view navigation dock:

Navigation View	DOM Container	Clinical & Operational Purpose
Ultra HRP	#view-uhrp	Primary command center with 8 risk factor tabs, NHM Hindi protocol box, triage KPI cards, and full 12-visit expandable drawer line list.
Moderate Anemia	#view-moderate	Dedicated triage for pregnant women with Hb 7.0–9.9 g/dL. Tracks parenteral iron initiation, 3rd trimester urgency, and delivery status.
Indicators	#view-indicators	District-wide visual charts: Management by risk/block, MO checkups, geographic block cases, and institutional delivery place breakdown.
All ANC	#view-anc	Total ANC registry line list with actionable KPI cards (Total Registered, High Risk, Delivery Overdue, Missed 2+ ANC, Not Seen by MO).

- **Desktop Navigation:** Persistent left sidebar with live counter badges and clean developer attribution.
- **Mobile Navigation:** Floating bottom pill navigation dock (blurred glass, rounded 36px) with touch-optimized targets.

6. Maintenance & Operational Guide for Health Administrators

To update data on a recurring monthly or weekly basis, health informatics officers follow this standardized workflow:

Step 1: File Ingestion

1. Download the latest Excel reports and place them into their respective folders: HRP line list/, ANC line list/, and Delivery linelist/.
2. Ensure new files follow the standard column format exported from the maternal tracking portal.

Step 2: Automated Pipeline Execution

1. Open PowerShell or Command Prompt in the project directory and execute: python update_dashboard.py
2. The script cleans raw records, merges deliveries, parses 12-visit longitudinal vitals, computes gestational age and delivery overdue metrics, outputs fresh data.js, and automatically recompiles this SOP documentation PDF.

Step 3: Verification & Deployment

1. Open index.html in any modern browser (Chrome, Edge, Safari, Firefox) to verify KPI counts and charts.
2. Push commits to GitHub repository to automatically publish updates to GitHub Pages.